**Series : AABB3/1**

**SET-3**

प्रश्न–पत्र कोड
Q.P. Code **56/1/3**

रोल नं.
**Roll No.**

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परीक्षार्थी प्रश्न–पत्र कोड को उत्तर–पुस्तिका के मुख–पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न–पत्र में मुद्रित पृष्ठ **12** हैं ।
- प्रश्न–पत्र में दाहिने हाथ की ओर दिए गए प्रश्न–पत्र कोड नम्बर को छात्र उत्तर–पुस्तिका के मुख–पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न–पत्र में **12** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर–पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न–पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न–पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न–पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर–पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।

- Please check that this question paper contains **12** printed pages.
- Q.P. Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **12** questions.
- **Please write down the Serial Number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

# रसायन विज्ञान (सैद्धांतिक)

# CHEMISTRY (Theory)

*निर्धारित समय : **2** घण्टे*
*Time allowed : **2** hours*

*अधिकतम अंक : **35***
*Maximum Marks : **35***

## General Instructions :

**Read the following instructions very carefully and strictly follow them.**

(i) *This question paper contains* ***12*** *questions.* ***All*** *questions are compulsory.*

(ii) *This question paper is divided into* **three** *Sections –* ***Section A, B*** *and* ***C.***

(iii) ***Section - A*** *Q. Nos.* ***1*** *to* ***3*** *are very short answer type questions carrying* ***2*** *marks each.*

(iv) ***Section - B*** *Q. Nos.* ***4*** *to* ***11*** *are short answer type questions carrying* ***3*** *marks each.*

(v) ***Section - C*** *Q. No.* ***12*** *is case based question carrying* ***5*** *marks.*

(vi) *Use of log tables and calculators is NOT allowed.*

### SECTION – A

1. Arrange the following compounds as directed : (any **Two**)

(i) In decreasing order of basic strength in aqueous solution :

$C_2H_5NH_2$, $(C_2H_5)_2$ NH, $(C_2H_5)_3$ N

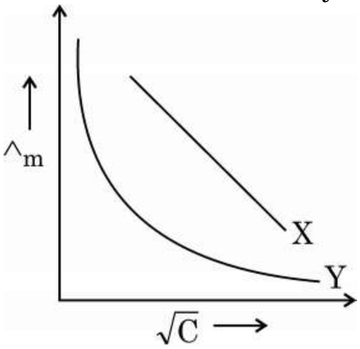
(ii) In increasing order of solubility in water :

$(C_2H_5)_2$ NH, $C_2H_5NH_2$, $C_6H_5NH_2$

(iii) In decreasing order of their pKb values :

$C_6H_5NH_2$, $C_2H_5NH_2$, $NH_3$ **1 × 2 = 2**

2. In the plot of molar conductivity ($\wedge_m$) Vs. square root of concentration $\left(\sqrt{C}\right)$, following curves are obtained for two electrolytes X and Y :




Answer the following :

(i) Predict the nature of electrolyte X and Y.

(ii) What happens on extrapolation of $\wedge_m$ to concentration approaching zero for electrolytes X and Y ? **1 × 2 = 2**

3. What do you mean by complex reactions ? Can we determine order and molecularity of a complex reaction ? **2**

## SECTION – B

4. (a) Write any three differences between physisorption and chemisorption. **1 × 3 = 3**

**OR**

(b) Define the following terms with a suitable example in each :

(i) Lyophilic sol

(ii) Macromolecular colloid

(iii) Coagulation **1 × 3 = 3**

5. (a) Illustrate the following reactions giving suitable example in each case :

(i) Gabriel phthalimide synthesis.

(ii) Carbylamine reaction.

(iii) Hoffmann bromamide degradation reaction. **1 × 3 = 3**

**OR**

(b) Write the structures of A, B and C in the following reactions :

(i) $CH_3CH_2Cl \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[0\ °C]{HNO_2} C$

(ii) $CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow[(b)\ H_2O]{(a)\ LiAlH_4} B \xrightarrow{C_6H_5SO_2Cl} C$ **1½ × 2 = 3**

6. An organic compound 'X' with the molecular formula $C_5H_{10}O$ forms 2,4-DNP derivative, does not reduce Tollens' reagent but gives positive iodoform test on heating with $I_2$ in the presence of NaOH. Compound 'X' gives ethanoic acid and propanoic acid on vigorous oxidation. Write the

(i) Structure of the compound 'X'.

(ii) Structure of the product obtained when compound 'X' reacts with 2,4-DNP reagent.

(iii) Structures of the products obtained when compound 'X' is heated with $I_2$ in the presence of NaOH. **1 × 3 = 3**

7. (a) (i) On the basis of crystal field theory, write the electronic configuration for $d^4$ ion if $\Delta_0 < P$.

(ii) Using valence bond theory, predict the hybridization and magnetic character of $[Ni(CN)_4]^{2-}$.

(Atomic number of Ni = 28)

(iii) Write the formula of the following complex using IUPAC norms :

Dichloridobis (ethane-1,2-diamine) cobalt (III) **1 × 3 = 3**

**OR**

(b) When a co-ordination compound $NiCl_2 \cdot 6H_2O$ mixed with $AgNO_3$, 2 moles of AgCl are precipitated per mole of the compound. Write

(i) Structural formula of the complex.

(ii) Secondary valency of 'Ni' in the complex.

(iii) IUPAC name of the complex. **1 × 3 = 3**

8. (a) Account for the following :

(i) Zr and Hf have almost similar atomic radii.

(ii) Transition metals show variable oxidation states.

(iii) Zinc has lowest enthalpy of atomisation. **1 × 3 = 3**

**OR**

(b) Why transition metals and their compounds act as good catalyst ? How is the variability in oxidation states of transition metals different from that of the non-transition metals ? Illustrate with examples. **3**

9. (a) The standard Gibbs energy ($\Delta rG°$) for the following cell reaction is $-300$ kJ mol$^{-1}$ :

$$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$$

Calculate $E^o_{cell}$ for the reaction. (Given: 1F = 96500 mol$^{-1}$)

(b) Calculate $\lambda^o_m$ for $MgCl_2$ if $\lambda^o$ values for $Mg^{2+}$ ion and $Cl^-$ ion are 106 S cm$^2$mol$^{-1}$ and 76.3 S cm$^2$mol$^{-1}$ respectively. **2 + 1 = 3**

10. Following ions of 3d transition series are given :

$Ti^{2+}$, $Fe^{2+}$, $Cu^{2+}$, $Zn^{2+}$

(Atomic number : Ti = 22, Fe = 26, Cu = 29, Zn = 30)

Identify the ion, which is

(i) a strong reducing agent in aqueous solution.

(ii) more stable than its +1 oxidation state.

(iii) colourless in aqueous solution.

Give a suitable reason in each. **1 × 3 = 3**

11. A first order reaction is 50% complete in 40 minutes. Calculate the time required for the completion of 90% of reaction.

[Given : log 2 = 0.3010, log 10 = 1] **3**

## SECTION – C

12. Read the passage given below and answer the questions that follow :

Aldehydes, ketones and carboxylic acids are some of the important classes of organic compounds containing carbonyl group. These are highly polar molecules due to higher electro-negativity of oxygen relative to carbon in the carbonyl group. Aldehydes are prepared by dehydrogenation or controlled oxidation of primary alcohols and controlled reduction of acyl halides. Ketones are prepared by oxidation of secondary alcohols and hydration of alkynes.

Aldehydes and ketones undergo nucleophilic addition reactions onto the carbonyl group but carboxylic acid does not undergo nucleophilic addition reaction. The alpha ($\alpha$) – hydrogens of aldehydes and ketones are acidic. Therefore aldehydes and ketones having at least one $\alpha$-hydrogen undergo Aldol condensation.

Aldehydes are easily oxidised by mild oxidising agents such as Tollens' reagent and Fehling's reagent. Carboxylic acids are prepared by the oxidation of primary alcohols, aldehydes and by hydrolysis of nitriles. Aromatic carboxylic acids are prepared by side-chain oxidation of alkyl benzenes. Carboxylic acids are considerably more acidic than alcohols and most of simple phenols.

(a) Arrange the following in the increasing order of their reactivity towards nucleophilic addition reaction. : **1**

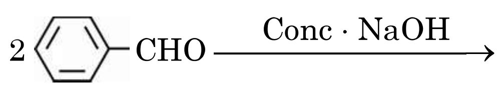
$CH_3COCH_3$, $CH_3CHO$, $HCHO$, $C_6H_5COCH_3$

(b) Give a simple chemical test to distinguish between Ethanal and Propanone. **1**

(c) Why carboxylic acid does not give nucleophilic addition reactions like aldehydes and ketones ? **1**

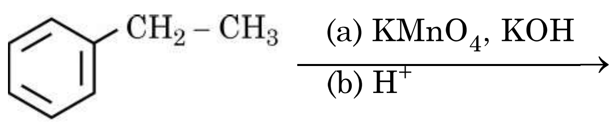
(d) (i) Why $\alpha$-hydrogen of aldehydes and ketones are acidic in nature ?

(ii) Write the products in the following : **1 + 1 = 2**

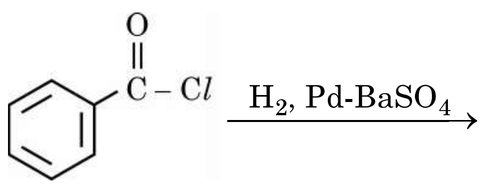



**OR**

Write the major products of the following reactions :

(i) 

**1**

(ii) 

**1**

**1+1+1+2=5**